Fiscal Year:	FY 2024	**Task Last Updated:**	FY 11/28/2023
PI Name:	Almeida, Eduardo A C Ph.D.		
Project Title:	The Role of P21/CDKN1a Pathway in Microgravity-Induced Bone Tissue Regenerative Arrest - A Spaceflight Study of Transgenic P21/CDKN1a Null Mice in Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Musculoskeletal Biology		
Space Biology Special Category:	None		
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Zip Code:	94035	**Congressional District:**	18
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Key Personnel Changes/Previous PI:	NA		
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Task Description:	Tissue degeneration and the failure of normal tissue regenerative processes in microgravity are increasing concerns for sustaining life in space as the duration of future flight missions increases. Specifically, our laboratory has identified normal adult stem cell-based tissue regenerative processes, such as the formation of new bone, cartilage, immune cells, and blood from bone marrow hematopoietic stem cell precursors, as being particularly sensitive to the stresses of spaceflight in microgravity. The cellular mechanisms activated in mouse bone during spaceflight are not completely understood, but they include induction of oxidative stress and the inhibition of differentiation of marrow mesenchymal stem cells, via activation of CDKN1a/p21-mediated cell cycle arrest in proliferative osteoprecursor cells on the bone surface. The identification of CDKN1a/p21 mRNA and protein as being highly up-regulated by spaceflight in mice is a seminal finding, because this gene is a master regulator of tissue regeneration, that when knocked out allows for mammalian limb regeneration similar to that found in amphibians. Because we established in multiple spaceflight and ground experiments that the CDKN1a/p21 pathway is a key molecular mechanism of inhibiting tissue regenerative processes specifically activated in microgravity, we sought to fly a CDKN1a/p21 null mice in microgravity to test the hypothesis that this gene is required for the arrest of regenerative bone formation observed in microgravity. Specifically, we hypothesized that CDKN1a/p21 is the key effector of cell cycle arrest in bone tissue response to the stresses of spaceflight in microgravity and more broadly that CDKN1a/p21 is required for bone regenerative arrest in mice during spaceflight. To test this hypothesis, we proposed the following specific goals: 1) To determine the extent of osteoblasts proliferation, differentiation, and mineral deposition in microgravity compared to 1g in a CDKN1a/p21-null background versus WT 2) To determine if the coupling and balance of increased osteoclastogenesis and decreased osteoblastogenesis in microgravity is altered in a CDKN1a/p21-null background 3) To determine if CDKN1a/p21 expression in microgravity is regulated via mRNA stabilization by the microgravity-induced Cold Induced RNA Binding Protein (CIRBP) in response to oxidative stress in bone. The experimental approaches originally proposed in 2014 to achieve these aims include MicroComputed tomography, bone histology, and histomorphometry of injected Calcein plus BromodeoxyUridine labels to determine bone growth, and bulk bone tissue genomics. We have however also recently updated our analysis plans to include novel methodologies that now allow us to quantify at the single cell level gene expression in the bone marrow stem cell niche containing various stages of differentiation of the osteoprogenitor and hematopoietic lineages. The combined results of these studies will allow us to definitively determine if the presence of CDKN1a/p21 during spaceflight induces arrest of bone growth and regeneration mediated by CDKN1a/p21 inhibition of osteoblastic proliferation and differentiation, and if so, how this might be coupled to osteoclastic activity induced in space. In addition, for the first time, these studies seek to connect the potential effector molecular mechanism of regenerative arrest in bone (by CDKN1a/p21) to its inducer, hypothesized here to be oxidative (and other) stress response gene CIRBP stabilization of CDKN1a/p21 message RNA. When completed, these studies will provide molecular mechanistic evidence for how CDKN1a/p21, one of the most important master regulatory genes of stem cell regenerative health, is modulated by microgravity and space radiation, and could be the basis for development of preventive treatments designed to counter tissue regenerative deficits in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Recent advances in the scientific understanding of stem cells and their role in tissue regeneration created a new field of regenerative medicine that holds great promise both in the treatment, as well as prevention, of many degenerative diseases often associated with aging. In short duration human spaceflight, when the body is weightless and unloaded in microgravity, multiple tissue degenerative conditions occur, such as muscle atrophy, bone loss, immune deficiencies, and others that may have a deficit in stem cell-based tissue regeneration in common. Likewise, on Earth, when we increase mechanical loading of the body with exercise we observe an increase in adult stem cell-based tissue regeneration, such as in bone strengthening, muscle mass increase, and increased production of immune cells and red blood cells from hemotopoeietic stem cell progenitors. In total, spaceflight and ground experimental evidence from human and animal studies now strongly suggests that stem cell-based regenerative health is regulated by mechanical loading of the body, increasing with exercise and load, and decreasing with inactivity and unloading, as in microgravity. Our proposed experimental work in microgravity flight experiments and mechanical unloading models addresses the molecular, cellular, and tissue aspects of the hypothesis that gravity mechanical loading regulates regenerative health. The novel results we obtained identify the transition between stem cell progenitors and differentiated cells as a likely cellular process regulated by mechanical loading and specific molecular candidates such as the p21/CDKN1a pathway induced in microgravity as inhibitory of stem cell-based tissue regeneration in bone. This work is highly significant to NASA space exploration and medicine on Earth because it holds the promise of greater scientific understanding of how to maintain and monitor health during future voyages of space exploration, and solidly establishes Earth gravity as a major shaper of human life through promoting tissue regenerative health.
	Progress in the current performance period included significant tissue sample processing for single cell (SC) genomics, plus microCt data acquisition and processing, as well as imaging of calcein tissue mineralization, as well as osteoclastic bone pitting by Scanning Electron Microscopy. Specifically: For single cell sequencing of bone marrow progenitors in femur mid-shafts dissected in orbit at the International Space Station (ISS) and on Earth for controls cryopreserved, we completed single cell mRNA barcoding with the 10X Genomics Chromium Controller, followed by Reverse Transcription of Chromium SC barcoded mRNAs, and generated, plus completed, sequencing mouse SC RR10. In particular, the second round of sequencing for 16 additional individual mouse bone marrow single cell mRNA samples was successfully processed in early 2023 to complete dataset gaps such as low cell counts in basal and other samples, plus to increase the N in the core flight (FLT), ground control (GC), p21/WT experiment from 3 to 6 mice, plus add 12 month aged samples for comparison to spaceflight. The 10X Genomics workflow from live cells to more than 800,000 final individual cell transcriptomes from 58 experimental RR10 samples is now fully integrated and normalized with the first round library preparation and sequencing performed in 2022; and analysis is underway, with preliminary results supporting rescue of osteogenesis by p21/Cdkn1a deletion and CIRBP involvement in p21/Cdkn1a over-expression. Because of the very large datasets involved, an ATLAS publication of the effects of spaceflight and related loading and unloading conditions (Hindlimb Unloading / HU, Running Wheel, Vivarium, Spaceflight, and Aging) in wild type (WT) mice is currently being prepared for publication first, to provide a background for the subsequent publication of the analysis of p21/Cdkn1a null mice spaceflight results. For Calcein incorporation and image analysis of mineralization,

Task Progress:	work is underway with confocal microscopy of optically cleared, mineralized whole mount femur shaft samples, with coupled Scanning Electron Microscopy imaging and analysis of osteoclastic pitting at matched bone sites. Preliminary results from a limited N suggest that RR10 WT mice show bone formation deficits as measured by Calcein injection labeling in space (75% reduction), while p21/Cdkn1a knockout in transgenic mice significantly rescues p21/Cdkn1a osteoblastic mineralization during the last week of spaceflight. Cdkn1a deletion, however, does not block microgravity-induced osteoclastic bone degradation. Also, while spaceflight reduces WT mineralizing surface area and cell number, it does not seem to affect mineral apposition rates in p21/Cdkn1a or WT mice at sites where mineralization from both injections overlap. MicroCT scanning is complete but analyses are still underway. Preliminary results already show that p21/Cdkn1a mice constitutively have lower bone mineral content than WT, likely due to high levels of osteogenesis coupled with osteoclastic degradation, with additional osteoclast-mediated bone loss in space, which may be stimulated by increased osteogenesis. Finally, while spaceflight in microgravity for 28 days during the Rodent Research-10 (RR-10) Mission in the NASA ISS Rodent Habitat appears to cause significant cortical and cancellous bone loss in load-bearing long bones such as the tibia and femur, other predominantly non-weight-bearing vertebrae show no significant cortical or cancellous bone loss. We conclude from this and consistent data on our ancillary studies of Rodent Research-1 (RR-1) archival NASA Ames Life Sciences Data Archive (ALSDA) specimens that bone loss in low-Earth orbit (LEO) is primarily due to weight-bearing, site-specific mechanical unloading in microgravity, and likely not from systemic effects/factors such as spaceflight stressors (including oxidative, hormonal, or radiation), which is an important finding that will allow us to more narrowly interpret the results with p21/Cdkn1a deletion as targeted to mechanotransduction and not other factors. In summary, RR-10 genomics data acquisition and microcomputed tomography (microCT) data scanning are complete at the end of 2023, but confocal and SEM, plus bromodeoxyuridine (BrdU) staining and imaging are still underway. Data analysis is well underway for genomics and computed tomography (CT) data but significant work remains, especially in the area of mineralization analysis. Progress on publishing RR-10 results include a manuscript submission in progress on: "Bulk Spinal Properties are Maintained After 28-Days of Spaceflight", Lim, S. Almeida, E. A. C., and O'Connell, G. D.; another manuscript submission on: "Cosmic Kidney Disease: The Effects of Spaceflight and Galactic Cosmic Radiation on Renal Structure and Function", Siew et al.; plus five American Society for Gravitational and Space Research 2023 (ASGSR 2023) meeting presentations and posters, as well as one Conference Paper presentation at the 2023 Summer Biomechanics, Bioengineering, and Biotransport Conference. [Ed. Note: See Bibliography for complete citations.]
Bibliography Type:	Description: (Last Updated: 12/08/2025)
Abstracts for Journals and Proceedings	Juran CM, Almeida EAC et al. "Spaceflight in microgravity regulates mouse bone marrow hematopoietic and mesenchymal tissue regenerative processes via a mechanism involving Cdkn1a." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstract ID #366. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023
Abstracts for Journals and Proceedings	Lim S, Almeida EAC. "Murine spine composition and mechanics are maintained after 30-Days of spaceflight." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023.Abstract ID #249. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023
Abstracts for Journals and Proceedings	Cahill R, Almeida EAC et al. "Deletion of Cdkn1a in mice rescues reduced osteogenesis in microgravity and increases coupled osteoclastic bone degenerative activity and bone loss." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstract ID #147. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023
Abstracts for Journals and Proceedings	Almeida EAC et al. "37-Day microgravity exposure in 16-week female C57BL/6J mice is associated with weight bearing site-specific bone loss." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstract ID #298. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023
Abstracts for Journals and Proceedings	Siew K, Almeida EAC et al. "Cosmic kidney disease: A pan-omic investigation of the health consequence of a trip to Mars and back." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstract ID #310. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023
Papers from Meeting Proceedings	Lim S, Veres JE, Almeida EAC, O'Connell GD. "Bulk properties of the murine spine are maintained during 30 days of microgravity on the International Space Station." Summer Biomechanics, Bioengineering, and Biotransport Conference,Vail, CO, June 4-8, 2023. Proceedings of the Summer Biomechanics, Bioengineering, and Biotransport Conference,Vail, CO, June 4-8, 2023. , Jun-2023